CREST AWARDS Student profile

First name *Please provide only the first name of the student/team member. Do not include middle or last names.*	Devadit
CREST Award Level	Gold
Project title	Behavioural Fingerprinting from Keystroke Dynamics
Mentor name *(if you had one)*	-

This Profile form is suitable for Bronze, Silver and Gold CREST Awards. Always use in conjunction with corresponding Student Guide Documents.

Please fill in this profile form and submit it along with your project work or report. This a requirement for your application and an opportunity for you to present clearly to the assessor how you have met the CREST criteria and reflected on your project.

If you are working in a group, make sure that you fill in and submit a separate Student Profile Form for each individual.

All scientists and engineers are creative. They use scientific and technical knowledge, make decisions, solve problems, evaluate and communicate their work. Throughout your CREST project you'll need to use and show these skills too.

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Criteria	Where do you show this in your report or project record?	Your notes to the assessor (optional)
1 – Planning your project		
1.1 You have set a clear aim for the project and have broken it down into smaller objectives	Section 1, page 2; Section 5, page 13; Section 8, page 18	The aim is one sentence on page 2, broken immediately below it into five numbered objectives, each ending in a "Done when" test. The last paragraph of Section 5 checks the objectives against the results, and Section 8 gives an objective-by-objective verdict: four met outright, the fifth (clustering) only in part, and the second half of the aim – beating the benchmark – not achieved.
1.2 You have explained a wider purpose for the project	Section 1, page 2; Section 8, pages 18–19	Page 2 opens with the break-in to my own Gmail account and then gives the scale of the problem, using Verizon's 2024 figure that about 88% of basic web-application attacks come from stolen passwords. Section 8 returns to that wider purpose and reports honestly on how much of it my result actually addresses.
1.3 You have identified a range of approaches to the project	Section 1, "Choosing a project approach", pages 2–4; Section 3, page 7	Five different ways of doing the whole project are identified and evaluated: (1) building and evaluating a working software system, (2) interviewing leading researchers in behavioural biometrics and keeping a video log, (3) producing a documentary or explanatory video, (4) writing a literature review and comparative analysis, and (5) running experiments with off-the-shelf tools without building anything. The table on page 4 evaluates each approach against my aim and against the five objectives, saying in each case what it would have produced, how effective it would be, which objectives it could and could not deliver, and the verdict with the reason. Section 3 covers the separate, later decision about the technical method, which is a different question from how the project as a whole would be done.

1.4 You have described your plan for the project and why you chose that approach	Section 1, pages 4–6; Section 3, pages 7–11	Page 4 gives the reasoning behind the choice. Only two of the five approaches produce a measurement at all, and only building my own model could answer the question I had set, which was whether a learned fingerprint can beat hand-designed methods on their own dataset. Each rejected approach carries its own reason: I had neither the equipment nor the editing skill to make a video that would do the subject justice; I am a school student with no institutional introduction to the researchers I would have wanted to interview, and an interview would have told me how the question might be answered rather than answering it; a literature review would have restated other people's results. Two of the rejected approaches were kept inside the project instead of beside it, as Section 2 and as the Stage 3 baseline. Pages 4–6 then describe the plan that followed from that choice, and Section 3 sets out the technical design with a trade-off table for the three ways the recognition itself could work.

1.5 You have planned and organized your time well	Section 1, pages 4–6; Figure 1 (Gantt chart), page 5	The plan on page 5 was drawn up in November 2025, before any project code was written, and was built backwards from the submission deadline I set myself. I kept it in my notebook rather than in the repository, and the table is that plan set out properly. For each of the six stages it gives the tasks, the planned dates, the duration, the milestone and its deadline, the resources needed, and the constraints and dependencies. Figure 1 on page 5 is a Gantt chart of the plan against what actually happened, with the six milestones marked, the deadline I set myself shown, and the two-month exam and SAT pause that I blocked out in advance. Pages 6–7 explain both overruns: four weeks on Stage 2 and four on Stage 3, absorbed by the planned pause, and a further two weeks taken out of Stage 6 when the evaluation had to be rebuilt after the training/test leak. The actual dates come from the project's commit history and can be checked.
2 – Throughout your project		
2.1 You have made good use of the materials and people available	Acknowledgements, page 23; Appendix B, page 25; Appendix D, pages 29–30	The acknowledgements name the dataset authors, the papers whose methods I used, the beginner AI course that gave me the grounding, and the library documentation and community forums that stood in for the colleagues a lab would have provided. Appendix B lists the exact library versions, the dataset SHA-256, the hardware and the one-command rerun.
2.2 You have researched the background to the project and acknowledged your sources appropriately	Section 2, page 7; References, pages 23–24; Acknowledgements, page 23	My background reading is written as an argument rather than a list: the 2009 study and TypeNet in 2021 bracket the field under completely different conditions, a recent ACM survey traces the movement between them, and the untested point where the two meet is the gap my project sits in. Every claim is cited by number to the sixteen references.

3 – Finalising your project		
3.1 You have made logical conclusions and explained the implications for the wider world	Section 6, pages 14–17; Section 8, pages 18–19	Section 6 now works through the whole threat model rather than only the case my system handles. The table on page 15 sets seven attacks side by side – credential stuffing, offline hash cracking, online brute force and dictionary attacks, copy-paste of a stolen password, phishing, shoulder-surfing and session hijack – and asks in each case whether a human is typing at all. The honest conclusion drawn from that is that the automated attacks, which are the ones that break most passwords, lie outside anything typing rhythm can defend, and that a perfect keystroke verifier would still not stop an offline hash crack. Page 16 gives four alternative applications for the technique and explains which of them a 14.2% error rate is actually good enough for. Section 8 is a Conclusions section that answers the aim directly with the numbers, checks each of the five objectives, and states the implications for the wider world: what the method is not a defence against, where it is genuinely useful, and why the 37-fold spread in per-person error rate means it must never be a sole check.
3.2 You have explained how your actions and decisions affected the project's outcome	Section 4, page 12; Section 5, page 13; Section 9, page 21; Appendix C, page 29	Fixing the training/test leak changed every number in the report. Wiring the blended verifier into the evaluation is where the 10.2% comes from, since until then the method I was most pleased with was being measured nowhere. Appendix C shows why I kept my original training settings after the tuning "winner" did worse on the real test.

3.3 You have explained what you have learnt and reflected on what you could improve	Section 9, pages 19–23	Section 9 evaluates the project and my own skills separately. Pages 19–20 set out what I learnt about the science and the method, then evaluate my own skills before and after the project – Python and PyTorch, experimental design, statistics, full-stack engineering, estimating my own time and scientific writing – with the evidence for each and what is still weak. The honest finding is that the leak lived in the gap between being able to build a model and being able to design an experiment. Pages 20–21 evaluate each of the six project stages separately, on whether it did what it was for, what went well and what went badly, because the stages did not succeed or fail together. Pages 21–22 cover the decisions that changed the outcome and the four things I would do differently. Pages 22–23 set out what it would take to repeat or extend the project, with the cost of each item: a GPU and roughly a week of rented compute for a large free-text dataset, six to eight weeks and an ethics review for a consented enrolment study, a few days for a calibrated confidence score, two to three weeks for the session-hijacking experiment, and one reviewer allowed to doubt me early, which would have cost nothing and saved two weeks.
4- Project-wide criteria		
4.1 You have shown understanding of the science behind your project, appropriate to the Award level	Sections 2–3, pages 7–11; Appendix C, pages 27–29; Whole report	Section 2 explains how a biometric system is scored and why a single accuracy figure would be meaningless when it can fail in two opposite directions. Appendix C works through the mechanism behind each technique in my own words, including why the covariance estimate has to be shrunk when only about twelve enrolment samples exist in 128 dimensions.

4.2 You have made decisions to direct the project, taking account of ethical and safety issues	Section 7, pages 17–18	Ethics decisions, not boilerplate: the data is special-category under GDPR, so the reported model is trained only on public anonymised data; consent is opt-in and revocable; only the derived fingerprint is stored, never the typing; every failure asks for another factor and never admits by default. The 37-fold error-rate spread between people is reported as a disparate-impact risk found in my own results, and the dual-use problem is stated openly.
4.3 You have shown creative thinking	Section 3, page 7; Section 6, pages 16–17	The central idea is running classical distance statistics inside a learned fingerprint, which I arrived at by taking the strongest part of the 2009 benchmark and putting it somewhere it had not been used. On this dataset, tested only on strangers, that measurement did not exist before.
4.4 You have identified and overcome problems successfully	Section 4, page 12; full log in the repository	Four problems are given in full, with cause, diagnosis, fix and verified outcome: the training/test leak that made my first result worthless; a live crash caused by an unusually consistent typist whose threshold fell near zero; a data file that trained on all-zero timings because its columns were named differently from my test fixture; and the blended verifier that ran in the live product while being measured nowhere. The full log of all thirteen fixes is in the project repository.
4.5 You have explained your project clearly, in writing or conversation	Whole report; Appendix A, page 24	The body is written to be followed without a technical background, with one everyday comparison for each hard idea. Every figure is captioned and referred to before it appears, the glossary in Appendix A covers the terms, and all the technical depth is quarantined in the appendices so the argument never depends on it.

Personal reflections

Now that you've finished your project, use this space to add further thoughts on what you did and evaluate each stage of the project process. The CREST Award student guide gives an example of what to include. You can continue on a separate sheet if necessary and use diagrams or pictures if you want to.
I chose this project after someone broke into my Gmail account. Luckily, I found out about the break very soon so I could secure my account before any significant damage was done. I had switched off 2FA in my account which got me thinking how easily someone with a stolen password can break into an account. While playing a typing game the idea struck me that what if I could make a system that not only looks at the password entered but also the manner in which it is typed. My project produced a 14.2% primary and 10.2% ensemble EER on 16 unseen subjects. I could not beat the 9.6% published baseline which I was hoping to exceed but I was still proud of myself and I documented the entire process in detail. Making many mistakes along the way was one of the disadvantages of not doing this project with a mentor but I learned a lot about AI through various sources including YouTube videos not only on matter directly related to my project but other concepts in the field of AI too. Going forward I plan on collecting a small consented set of real users to test the model and then based on their feedback and typing results I can improve it accordingly. I believe if I can tune this model to eventually not just cross the 9.6% EER baseline but far surpass it, the application for this project is endless and something like this truly has the potential to change the world of cybersecurity.
My mentor or supervisor
If you had a project mentor or placement supervisor (such as a teacher, relative, industry expert, etc.) how did they help with your project? (Please leave blank if you did not have one.)
-
Ask your mentor to confirm that this project is your work by signing below. **Signature of mentor or supervisor**
Date: 30/08/2026

Space for further notes/ drawings/ reflections (optional)

The system is live at 'typing-sanctuary.onrender.com'. The Train page builds a typing profile in the browser and shows a similarity score when you type again, so you can watch the idea work in about two minutes. It runs on a free hosting tier that sleeps when idle, so the first page load can take up to twenty seconds; it is not broken, only waking up.

Everything in the report can be re-run locally. The code, data pipeline and the full log of thirteen fixes are publicly available at 'github.com/devadit1515/typing-sanctuary'.

How I used AI. My use of AI is declared in full in the report (A Note on AI use, page 23). In short: I used Anthropic's Claude as a coding aid and to think ideas through across the project, and again when revising this report in August 2026. It drafted some functions and helped me track down bugs; I reviewed, ran and tested everything it produced, and the commit history holds the dated trail.

Note on this resubmission. This is a revised submission, and the changes below were made in response to the assessor's feedback.

1.3 and 1.4 – Section 1 now identifies five different approaches to the project as a whole, not only to the software: building and evaluating a working system, interviewing leading researchers with a video log, producing a documentary, writing a literature review, and running experiments with off-the-shelf tools. A table on page 4 evaluates each one against my aim and my five objectives, and page 4 gives my reasons for choosing to build and for rejecting each alternative.

1.5 – The retrospective summary has been replaced by the plan I drew up in November 2025, before any project code, kept in my notebook and tracked against the commit history. It gives tasks, timescales, resources, constraints, dependencies, milestones and deadlines for all six stages, and Figure 1 on page 5 is a Gantt chart of that plan against what actually happened, with an explanation of both overruns and how I replanned.

3.1 – Section 6 now covers copy-paste and automated password cracking, with a table on page 15 comparing seven attacks and asking in each case whether a human is typing at all. A new Conclusions section (Section 8) answers the aim directly, checks each objective, and sets out where the technique is and is not useful in the wider world.

3.3 – Section 9 now evaluates my own skills before and after the project, evaluates each of the six stages separately, and states the time, hardware, funding and ethical approval that repeating or extending the project would require.